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Beyond Boundaries

Agentic Ai Lab

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Working of 5 Levels of Text Splitting

Introduction

Text splitting (also called chunking) is the process of breaking large text into smaller, manageable pieces so that language models can process, store, and retrieve information effectively. It is a **core concept in RAG (Retrieval-Augmented Generation), search systems, agents, and long-context applications.**

This document explains **how each level of text splitting works**, why it exists, and **when to use it**, without repeating any code.

Level 1: Character Splitting

How it Works

The text is divided purely based on a **fixed number of characters**. The splitter does not understand sentences, words, or meaning. It simply counts characters and cuts the text accordingly.

Optionally, a small overlap is kept between chunks to avoid losing information at boundaries.

Key Idea

- Splitting is mechanical
- No understanding of language structure

Strengths

- Very easy to implement
- Fast and predictable

Weaknesses

- Breaks sentences and ideas
- Loses semantic meaning
- Poor retrieval quality

When to Use

- Learning chunking concepts

- Very rough preprocessing
 - Never recommended for production RAG systems
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Level 2: Recursive Character Splitting

How it Works

Instead of blindly cutting text, recursive splitting tries **multiple separators in order**:

1. Paragraph breaks
2. Line breaks
3. Spaces
4. Characters (last resort)

The splitter attempts to keep chunks within size limits **while respecting text structure as much as possible**.

Key Idea

- Structure-aware splitting
- Falls back gracefully if ideal splits are not possible

Strengths

- Preserves sentences and paragraphs
- Much better context retention
- Works well for most text types

Weaknesses

- Still not meaning-aware
- Cannot understand topic shifts

When to Use

- Default choice for most RAG applications
 - Blogs, articles, documentation, books
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Level 3: Document-Specific Splitting

How it Works

This level uses **custom splitters depending on document type**, such as:

- Python code split by functions and classes
- Markdown split by headers
- PDFs split by pages or sections

Each splitter understands the **structure rules of that document format**.

Key Idea

- One document type $\neq$ another
- Structure matters more than size

Strengths

- Highly accurate chunk boundaries
- Preserves logical sections
- Excellent for code and technical docs

Weaknesses

- Requires correct document detection
- More setup than generic splitters

When to Use

- Codebases
- Markdown documentation
- PDFs, research papers, manuals

Level 4: Semantic Splitting

How it Works

Instead of using characters or separators, semantic splitting uses **embeddings**:

- Each sentence is converted into a vector
- Similar sentences are grouped together
- Topic changes trigger new chunks

This ensures each chunk contains **one coherent idea**.

Key Idea

- Meaning over structure
- Similarity-based grouping

Strengths

- Best contextual coherence
- Excellent retrieval accuracy
- Ideal for complex knowledge bases

Weaknesses

- Computationally expensive
- Slower preprocessing
- Requires embedding models

When to Use

- High-quality RAG systems
- Knowledge assistants
- Enterprise search

Level 5: Agentic Splitting

How it Works

An agent (rule-based or LLM-powered) decides **how and where to split text** based on intent, goals, or reasoning.

The agent may:

- Detect topic changes
- Identify important entities
- Decide chunk boundaries dynamically

Key Idea

- Splitting is a decision-making process
- Context-aware and goal-driven

Strengths

- Extremely flexible
- Adapts to task requirements
- Best for autonomous systems

Weaknesses

- Experimental
- Hard to debug
- Expensive and complex

When to Use

- AI agents
- Autonomous research systems
- Advanced multi-step RAG pipelines

Comparison Summary

	Level	Awareness	Accuracy	Cost	Typical Use
1	None	Very Low	Very Low	Very Low	Learning only
2	Structure	Good	Low	Low	Default RAG
3	Format	High	Medium	Medium	Code, PDFs
4	Meaning	Very High	High	High	Production RAG
5	Reasoning	Highest	Very High	Very High	AI Agents

Final Takeaway

- **Start with Level 2** for most applications
- **Upgrade to Level 3** when document structure matters
- **Use Level 4** when accuracy is critical
- **Level 5** is for future-ready, agentic systems

Text splitting is not about breaking text — it is about **preserving meaning while respecting model limits**.